

# SIS SAMPLE DRAFT — FULL TIER

## Sacramento County Sample

---



### *Traffic Impact Study*

Prepared September 2, 2026 · 38.7016, -121.2733

**SIS sample draft — full tier**

# Traffic Impact Study

Generated 2026-09-02T23:03:04.676Z

## EXECUTIVE SUMMARY

This report addresses the transportation impacts associated with the proposed Sacramento County Sample located within Sacramento-Roseville-Folsom MSA, California. The host lead agency is Caltrans + OPR Dec 2018 defaults; the regional MPO is Regional MPO covering project site.

### SB 743 FRAMING — SCOPE OF THIS REPORT

California Senate Bill 743 (Stats. 2013, Ch. 386), codified at Public Resources Code § 21099(b)(2) and implemented through CEQA Guidelines 14 CCR § 15064.3 (effective statewide 2020-07-01), replaced LOS-based CEQA transportation impact analysis with Vehicle Miles Traveled (VMT) analysis. Per § 21099(b)(2), "automobile delay, as described solely by level of service or similar measures of vehicular capacity or traffic congestion, shall not be considered a significant impact on the environment."

This report presents §4 non-CEQA operational LOS analysis suitable for Caltrans District Encroachment Permit review (SHS frontage); local agency site-access design review, AND §3 a Tier-1 CEQA-VMT screening determination that auto-evaluates the six OPR § E.1 screening criteria against project metadata. The §4 LOS analysis does NOT, by itself, satisfy CEQA transportation-impact requirements. If §3 auto-screens the project out, the project is presumed less-than-significant under CEQA Guidelines § 15064.3 without further VMT analysis; otherwise, §3 surfaces the inputs needed for a full VMT determination under the OPR (LCI) Technical Advisory on Evaluating Transportation Impacts in CEQA (Dec 2018); local lead-agency adoption status to be confirmed methodology (OPR Technical Advisory, Dec 2018).

Findings (operational scope):

- 7 intersections are projected to drop one or more LOS grades under build conditions (non-CEQA scope).
- 2 intersections operate at LOS E or F under build conditions and may warrant operational mitigation (non-CEQA scope).

|                            |                       |                        |                               |
|----------------------------|-----------------------|------------------------|-------------------------------|
| <b>10</b><br>INTERSECTIONS | <b>7</b><br>LOS DROPS | <b>2</b><br>AT LOS E/F | <b>15.4s</b><br>WORST Δ DELAY |
|----------------------------|-----------------------|------------------------|-------------------------------|

## 1.0 PROJECT DESCRIPTION

### 1.1 Project Location

The proposed Sacramento County Sample is located within Sacramento-Roseville-Folsom MSA, California. The host CEQA lead agency for the transportation determination is Caltrans + OPR Dec 2018 defaults.

### 1.2 Project Summary

|                  |                                    |
|------------------|------------------------------------|
| Project name     | Sacramento County Sample           |
| Land use         | LU 820 — Shopping Center / Retail  |
| Development size | 85 1,000 sqft GLA                  |
| Site coordinates | 38.7016°, -121.2733°               |
| Opening year     | 2027                               |
| Host lead agency | Caltrans + OPR Dec 2018 defaults   |
| Regional MPO     | Regional MPO covering project site |

### 1.3 Site Access and Multimodal Context

Driveway-level ingress/egress, internal circulation, bicycle/pedestrian network connectivity, and transit access detail are dependent on the final site plan and are not produced by this screening tool. Site-plan-stage analysis is recommended for any formal submittal.

## 2.0 REGULATORY FRAMEWORK

## 2.1 CEQA / SB 743 / OPR Technical Advisory

CEQA transportation-impact significance for this project is governed by Pub. Resources Code § 21099 (SB 743, Stats. 2013, Ch. 386), CEQA Guidelines 14 CCR § 15064.3 (adopted Dec 28, 2018; effective statewide 2020-07-01), and the OPR (now Governor's Office of Land Use and Climate Innovation, LCI) Technical Advisory on Evaluating Transportation Impacts in CEQA (Dec 2018). Per § 15064.3(a), VMT is the default transportation metric for CEQA impact significance determinations.

## 2.2 Local Lead Agency Guidelines — Caltrans + OPR Dec 2018 defaults

The applicable local lead-agency methodology and significance thresholds derive from: OPR (LCI) Technical Advisory on Evaluating Transportation Impacts in CEQA (Dec 2018); local lead-agency adoption status to be confirmed. Significance threshold: 15% below the Regional MPO VMT/capita and VMT/employee (OPR default); county-level if region is much larger than commute-shed (OPR p. 16). Screening floor: <110 daily project trips presumed less-than-significant. Project-level VMT estimation source: Regional MPO covering project site travel demand model or jurisdiction-published calculator (host agency to confirm).

Note. No host-jurisdiction-specific TIA guidelines were identified for this site. Confirm the local lead agency's adopted VMT guidelines and threshold before submittal — most California jurisdictions adopted between 2019 and 2024 and many post-date OPR defaults.

## 2.3 Caltrans Authority on State Highway System Frontage

Where the project fronts on a Caltrans-owned State Highway System (SHS) facility, the Caltrans Transportation Analysis under CEQA (TAC, 2nd Edition, Sept 2025) governs CEQA significance determination for SHS impacts and invokes the Caltrans Transportation Analysis Framework (TAF, 2nd Edition, Sept 2025) for induced-travel analysis on capacity-increasing SHS projects. Non-CEQA encroachment permitting in state right-of-way is governed by the Caltrans Encroachment Permits Manual (EPM); HDM-compliant operational analysis (LOS, queueing, signal warrants per CA MUTCD 2026 Part 4C) is expected for any new access onto an SHS facility.

## 3.0 CEQA-VMT IMPACT DETERMINATION

### TIER-1 SCREENING DETERMINATION (Phase 2)

This section auto-evaluates the six OPR § E.1 screening criteria against project metadata where the engine has the inputs and surfaces the remaining GIS-dependent criteria (TPA / low-VMT TAZ / redevelopment baseline) as "Requires verification." If any criterion auto-screens out, the project is presumed less-than-significant under CEQA Guidelines § 15064.3 without further VMT analysis. Otherwise, §3.1 / §3.2 list the baseline + project-VMT inputs needed to apply the §3.4 significance threshold; the auto-screening engine does NOT fabricate VMT numbers or substitute for a regional MPO model run (see REGIONAL-SPECS/california-vmt-spec.md §5 for Tier-1 scope and §3.6 below for Tier-2 wiring roadmap).

### 3.1 Required Baseline VMT Inputs

|                                              |                                                                                                                                       |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Baseline geography                           | Regional MPO VMT/capita and VMT/employee (OPR default); county-level if region is much larger than commute-shed (OPR p. 16)           |
| Residential metric (OPR Tech Advisory p. 10) | Home-based VMT per capita (tour-based ideal; trip-based acceptable)                                                                   |
| Office / employment metric (OPR p. 16)       | Home-based work VMT per employee                                                                                                      |
| Retail metric (OPR p. 16)                    | Net change in total VMT (absolute, not per-capita)                                                                                    |
| MPO travel-demand model                      | Host MPO model — confirm at scoping                                                                                                   |
| Current RTP/SCS                              | Host MPO RTP/SCS — confirm at scoping                                                                                                 |
| MPO baseline portal / source                 | <a href="https://dot.ca.gov/programs/sustainability/sb-743/resources">https://dot.ca.gov/programs/sustainability/sb-743/resources</a> |
| Published baseline status                    | No host-jurisdiction baseline lookup; use OPR floor (15% below regional baseline) and pull baseline from host MPO RTP/SCS at scoping. |
| Optional jurisdiction calculator             | None published — MPO model run required                                                                                               |

### 3.2 Required Project VMT Inputs

Per OPR Technical Advisory p. 4-5, the project VMT estimate and the threshold must use the same

method (apples-to-apples constraint — tour-based with tour-based, trip-based with trip-based).

|                                    |                                                                                                                     |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Required project VMT estimate      | Regional MPO covering project site with-project travel demand model run, or the jurisdiction's published calculator |
| Required method consistency        | Project method MUST match baseline method (tour-based OR trip-based; not mixed)                                     |
| Required cumulative scenario       | Project + reasonably-foreseeable cumulative projects vs. RTP/SCS horizon year baseline                              |
| Required RTP/SCS consistency check | Project alignment with Regional MPO covering project site Sustainable Communities Strategy                          |

### 3.3 Auto-Screening Cascade (OPR § E.1, p. 12-14)

The six OPR screening criteria below are auto-evaluated against project metadata where the engine has the inputs (daily trip count, land-use category, project size). Criteria that require GIS layers the engine does not yet ingest (Transit Priority Area, low-VMT TAZ map, prior-use VMT for redevelopment) are flagged "Requires verification" with the data source named. If ANY criterion resolves to "Screened out," the project is presumed less-than-significant for CEQA-VMT purposes and a full VMT impact analysis is not required.

| OPR Criterion                                                                                                                   | Auto-screening result                           | Notes                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Small project: <110 daily trips (Caltrans + OPR Dec 2018 defaults screening threshold; OPR floor 110)                           | Not screened out by this criterion              | Project generates 6,800 daily trips. Threshold: 110.                                                                                                                                                                                                                                                                                                   |
| Transit Priority Area (TPA): within ½ mi of a major transit stop (PRC § 21064.3) or high-quality transit corridor (PRC § 21155) | Requires verification (data source named below) | Requires GIS query against the MPO's major-transit-stop layer + high-quality-transit-corridor layer. TPA presumption does NOT apply if FAR <0.75, parking exceeds requirement, project is inconsistent with the SCS, or affordable units are replaced with fewer market-rate units (OPR Tech Advisory p. 14) — flag in submittal even if TPA-eligible. |
| Low-VMT area: project sited in a TAZ already performing ≥15% below baseline                                                     | Requires verification (data source named below) | Consult the host jurisdiction's published low-VMT screening map (e.g., SCAG HELPR 3.0; SANDAG SB 743 portal; LADOT VMT Calculator zone lookup; Fresno COG screening tool). Auto-screening from project lat/lon not implemented in this Phase-2 slice.                                                                                                  |
| Locally-serving retail <50,000 sf                                                                                               | Requires verification (data source named below) | Project is land use 820 (retail) but size unit (1,000 sqft GLA) is not in ksf; cannot auto-compare. Convert to ksf and reapply.                                                                                                                                                                                                                        |
| 100% affordable residential infill                                                                                              | N/A for this project                            | Project is land use 820, not residential.                                                                                                                                                                                                                                                                                                              |

| OPR Criterion                                                     | Auto-screening result                           | Notes                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Redevelopment with net VMT decrease (existing use → proposed use) | Requires verification (data source named below) | Requires prior-use VMT computation (existing site land use + intensity + tenancy). If the site is vacant or undeveloped, this criterion does not apply — flag as N/A in submittal. OPR p. 14: presumption does not apply where redevelopment displaces affordable housing near transit. |

**AUTO-SCREENING RESULT: NOT screened out by any auto-evaluable criterion. Verification-pending criteria above may still resolve to screened-out; otherwise, complete the §3.1 / §3.2 baseline + project-VMT inputs and apply the §3.4 significance threshold.**

Note: this screening cascade does not fabricate VMT numbers or substitute for a full MPO model run. The TPA, low-VMT-map, and redevelopment criteria are GIS-dependent and on the Phase-2 roadmap (per REGIONAL-SPECS/california-vmt-spec.md § 5).

### 3.4 Significance Threshold

Per OPR (LCI) Technical Advisory on Evaluating Transportation Impacts in CEQA (Dec 2018); local lead-agency adoption status to be confirmed: project VMT (per capita for residential, per employee for office, total for retail) is significant if it exceeds 85.0% of the Regional MPO VMT/capita and VMT/employee (OPR default); county-level if region is much larger than commute-shed (OPR p. 16) (i.e., reduction of less than 15% from baseline). The baseline value must be drawn from the Regional MPO covering project site latest published RTP/SCS travel demand model; the value is not hardcoded in this report because MPO model updates and RTP/SCS cycles shift the published number.

### 3.5 VMT-Reduction Mitigation Menu (CAPCOA 2024)

If the project exceeds the §3.4 significance threshold, the CEQA-VMT analysis must propose VMT-reducing mitigation drawn from the CAPCOA Handbook for Analyzing GHG Emission Reductions, Assessing Climate Vulnerabilities, and Advancing Health and Equity (2024 Edition, adopted 2024-11-21; supersedes Dec 2021). The categories and representative measures below are the Tier-1 reference menu. Project-specific reduction percentages must be computed in CAPCOA's per-measure formulas, parameterized on context (urban / suburban / rural; transit availability), with the multiplicative stacked-measures cap applied to prevent double-counting.

| CAPCOA category     | Representative measures                                                                                                                                                             | Reduction range (context-dependent)                                                                  |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Land use            | Increased residential density (T-1); increased job density (T-2); diverse / mixed-use land use (T-3); affordable-housing integration (T-4)                                          | Up to ~30% (density); ~9% (mixed-use); ~10% (affordable inclusion). Urban context, project-specific. |
| Neighborhood design | Improved pedestrian network (T-7); traffic calming (T-9, T-26); local-serving retail (T-27); intersection-density / street-grid improvements                                        | 0~7% per measure; high-leverage when combined with transit access.                                   |
| Transit             | Transit-accessibility improvements (T-5); subsidized / discounted transit (T-12); bicycle end-trip facilities (T-13); bike-sharing (T-29); EV charging (T-31)                       | 0~15% (transit access); transit subsidy ~0.3-20% per CAPCOA Ch. 3.                                   |
| Parking management  | Limit residential parking supply (T-22); price residential parking (T-23); unbundle parking costs (T-24); price workplace parking (T-20); cash-out (T-21); car-sharing (T-18, T-28) | Parking supply / pricing among the highest-leverage measures in suburban contexts.                   |

| CAPCOA category           | Representative measures                                                                                                                                                                                       | Reduction range (context-dependent)                                                                   |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Trip reduction / TDM      | Workplace TDM program (T-10, T-30); ride-share (T-11); marketing (T-14); compressed work week (T-15); telecommute / WFH (T-16); vehicle-trip caps (T-17); school-pool (T-19)                                  | Workplace TDM up to ~21%; telecommute / compressed weeks scale with eligible employee share.          |
| Pricing / road management | Cordon / area pricing; HOT lanes; VMT fee programs; locally-administered VMT mitigation funds (e.g., San Diego Mobility Choices Active Transportation In-Lieu Fee, City of Fresno VMT Reduction Program 2025) | Context-specific; programmatic mitigation typically requires nexus / fee adoption by the lead agency. |

Measure codes (T-1 ... T-31) reference CAPCOA 2024 Handbook Ch. 3. Reduction ranges above are illustrative ceilings drawn from the published handbook tables; the binding project-specific reduction is computed from the per-measure formula in the host jurisdiction's adopted CAPCOA edition (some jurisdictions still lock to CAPCOA Dec 2021 — confirm at scoping). The 2024 update split measures into (a) sufficient-evidence-for-quantified-reductions and (b) requires-additional-evidence; only (a) is creditable as primary mitigation, (b) is supplemental.

Important: intersection geometry improvements (turn lanes, signal-timing retiming, queue jumps, additional through lanes) do NOT reduce VMT and are NOT creditable mitigation under CEQA Guidelines § 15064.3(b)(1). Those improvements remain creditable as non-CEQA operational improvements (see §4 of this report) but cannot serve as CEQA-VMT mitigation. Caltrans' July 2022 Mitigation Playbook (citing CAPCOA Dec 2021, now superseded by 2024) specifically endorses increased density and affordable-housing inclusion as the highest-leverage residential VMT mitigations.

### 3.6 Cumulative VMT Determination

Per CEQA Guidelines § 15130 and OPR Technical Advisory p. 17, the CEQA-VMT analysis must evaluate the project's contribution to cumulative VMT impacts in the host RTP/SCS horizon year (Host MPO RTP/SCS — confirm at scoping). Two cumulative scenarios are required: (1) cumulative no-project — the RTP/SCS horizon-year baseline VMT including all reasonably-foreseeable cumulative projects; (2) cumulative plus project — the same horizon plus this project's VMT. A project's cumulative contribution is significant when the project's incremental contribution exceeds the \$3.4 threshold OR when the cumulative no-project scenario already exceeds the regional GHG-reduction target and the project is not RTP/SCS-consistent (OPR p. 17).

This screening tool does not auto-generate cumulative VMT — the cumulative scenario requires an MPO horizon-year model run with the reasonably-foreseeable project pipeline coded. The inputs below identify what the consultant must compile.

|                                         |                                                                                                                                                                                                       |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Horizon year                            | Host MPO RTP/SCS — confirm at scoping                                                                                                                                                                 |
| Cumulative no-project source            | Regional MPO covering project site Host MPO model — confirm at scoping run at the RTP/SCS horizon year, including all reasonably-foreseeable cumulative projects per CEQA Guidelines § 15130(b)(1)(B) |
| Cumulative plus-project source          | Same MPO model run with this project coded into the cumulative pipeline                                                                                                                               |
| Reasonably-foreseeable project pipeline | Pulled from host jurisdiction's pending-applications register and the MPO's RTP committed-projects list                                                                                               |
| GHG-reduction-target benchmark          | CARB SB 375 regional GHG target for the host MPO; cumulative exceedance + project inconsistency triggers § 15130 significance                                                                         |
| RTP/SCS consistency check               | See §3.7                                                                                                                                                                                              |

### 3.7 RTP/SCS Consistency Narrative

Per CEQA Guidelines § 15125(d) and § 15183.3, the project must demonstrate consistency with the host MPO's Sustainable Communities Strategy (Host MPO RTP/SCS — confirm at scoping). A project that is RTP/SCS-consistent receives presumption that its land-use pattern aligns with the regional GHG-reduction trajectory. Inconsistency does not by itself render a project significant, but it removes the consistency presumption and shifts the burden onto the lead agency to demonstrate that the project's deviation from the SCS does not cumulatively undermine the SB 375 target.



|                                             |                                                                                                                                                                                                                                                         |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCS land-use designation                    | Pull the project parcel's SCS designation from the Regional MPO covering project site RTP/SCS land-use overlay                                                                                                                                          |
| Project land use vs. SCS designation        | Apply the SCS-consistency test from the host MPO's adopted SCS consistency review framework (varies by MPO — SCAG uses the SCS Consistency Review Process; MTC uses the PBA 2050 Implementation Plan; SANDAG uses the SB 375 Consistency Determination) |
| Transit Priority Project (TPP) eligibility  | PRC § 21155 — TPPs receive streamlined CEQA review; check if project qualifies                                                                                                                                                                          |
| Priority Development Area (PDA) eligibility | MTC region: PBA 2050 PDA overlay confers SCS-consistency presumption                                                                                                                                                                                    |
| Priority Conservation Area (PCA) overlap    | Project sited in a PCA cannot claim SCS consistency without explicit lead-agency variance                                                                                                                                                               |
| GHG-reduction policy alignment              | Project's role in helping the host MPO meet its CARB SB 375 GHG target (per the latest CARB Sustainable Communities Strategy Evaluation)                                                                                                                |

Note: SCS consistency is a narrative determination by the lead agency, not a computed metric. The renderer surfaces the required inputs; the narrative itself is compiled by the consultant against the host MPO's consistency review framework prior to submittal.

## 4.0 NON-CEQA OPERATIONAL ANALYSIS

### SCOPE — NON-CEQA OPERATIONAL ANALYSIS ONLY

Per Pub. Resources Code § 21099(b)(2) and CEQA Guidelines § 15064.3(a), level-of-service analysis does not constitute a CEQA transportation-impact determination. This section is provided for non-CEQA operational review including Caltrans District Encroachment Permit review (SHS frontage); local agency site-access design review — that is, Caltrans Encroachment Permit review under the Encroachment Permits Manual, signal warrant analysis under CA MUTCD 2026 Part 4C, queueing and site-access design under Highway Design Manual Chapters 100 and 400, and the local agency's adopted operational standards.

#### 4.1 Methodology

Level of Service (LOS) is calculated from average control delay per vehicle: Webster's uniform-delay equation (F. V. Webster, Traffic Signal Settings, Road Research Technical Paper No. 39, Road Research Laboratory / HMSO, London, 1958) plus Akcelik's time-dependent overflow term, with a standard undersaturated cyclic back-of-queue relation at the 95th percentile. Control-delay bands (as republished in state DOT traffic-engineering manuals):  $A \leq 10s$  ·  $B \leq 20s$  ·  $C \leq 35s$  ·  $D \leq 55s$  ·  $E \leq 80s$  ·  $F > 80s$  of average control delay per vehicle. Caltrans Highway Design Manual (HDM, 7th Edition) Topic 102 + Ch. 400 governs LOS-based design capacity for SHS facilities. CA MUTCD 2026 Part 4C governs signal-warrant analyses.

#### 4.2 Trip Generation

Gross trip generation is calculated per the public-data screening rates (NHTS 2017 / SANDAG 2002 / NCHRP 716) for land use 820 (Shopping Center / Retail) at the proposed development size of 85 1,000 sqft GLA. Background traffic growth is applied at 1.50% per year, applied as an explicit override, not derived from measured data. Explicit override — 1.50%/yr was supplied with the request and applied in place of the engine's own rate. No measured growth rate is wired for this region, so there is no measured value to compare the override against. The applied rate is not derived from measured data; its basis must be stated by the preparer. Pass-by capture applied: 30%; internal capture applied: 0%.

| Period       | Entering trips | Exiting trips |
|--------------|----------------|---------------|
| Daily        | 3,400          | 3,400         |
| AM peak hour | 166            | 106           |

| Period       | Entering trips | Exiting trips |
|--------------|----------------|---------------|
| PM peak hour | 326            | 354           |

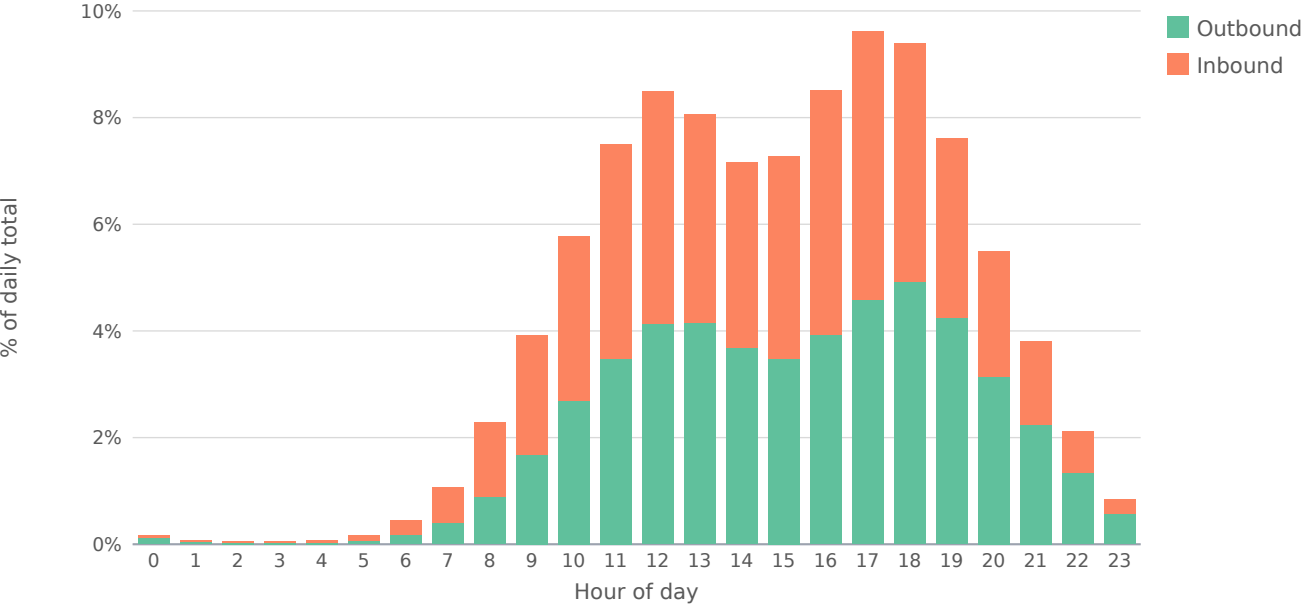
| Period          | Raw   | Pass-by | Int. cap. | External | In    | Out   |
|-----------------|-------|---------|-----------|----------|-------|-------|
| AM Peak Hour    | 272   | 20      | 0         | 225      | 137   | 88    |
| PM Peak Hour    | 680   | 204     | 0         | 426      | 204   | 222   |
| Saturday Midday | 748   | 56      | 0         | 619      | 309   | 310   |
| Daily Total     | 6,800 | 510     | 0         | 5,626    | 2,813 | 2,813 |

The external / net-external column is net-new vehicle trips assigned to the roadway = (gross – pass-by – internal-capture) × auto-mode share (89.3% for this metro). Walk / bike / transit person-trips are excluded from off-site assignment, so In + Out equals the external column.

### Trip Distribution by Time of Day

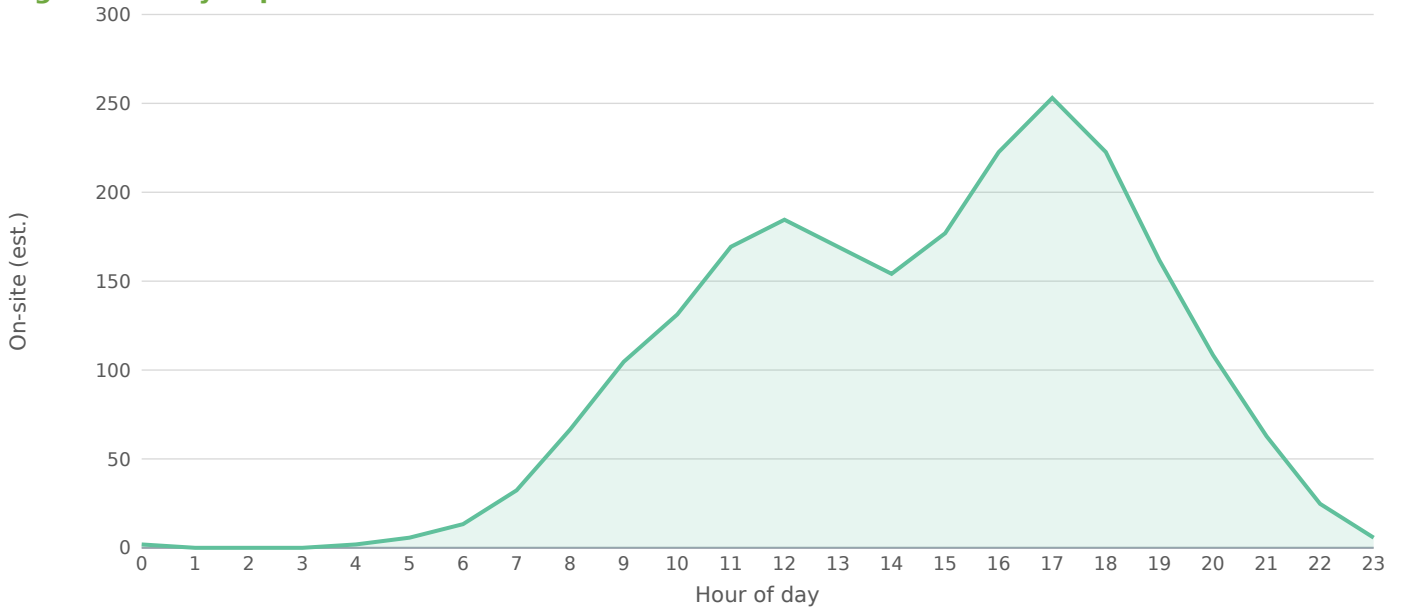
Estimated within-day distribution of the 6,800 gross daily trips and the resulting on-site accumulation. Within-day distribution from NHTS 2022 shopping / errand / meal trip start times, cross-checked against public-data time-of-day patterns for LU 820 / 850. Activity builds to a midday plateau and a stronger late-afternoon/evening peak; on-site occupancy peaks midday through the evening.

**Figure — Inbound / Outbound Trips by Start Time**





**Figure — Daily Trip Accumulation**



Peak on-site accumulation ~253 at 17:00, from 6,800 gross daily trips on the retail within-day profile. Screening estimate; not a substitute for a calibrated time-of-day model.

### 4.3 Trip Distribution — Gravity Model

The trip distribution uses the gravity model (term =  $M / (d^1 \cdot d_{site})$ ; mass basis: intersection through-volume (AADT × K-factor) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

| Directional sector    |  | Share of project trips |  |  |  |
|-----------------------|--|------------------------|--|--|--|
| Northeast (NNE + ENE) |  | 49%                    |  |  |  |
| Southeast (ESE + SSE) |  | 41%                    |  |  |  |
| Southwest (SSW + WSW) |  | 4%                     |  |  |  |
| Northwest (WNW + NNW) |  | 6%                     |  |  |  |

| Study-area zone                       | Dir. | Distance (mi) | Mass (M) | Gravity term | Trip share |
|---------------------------------------|------|---------------|----------|--------------|------------|
| Sunrise Boulevard & Oak Avenue        | SSE  | 0.11          | 1,529    | 0.1          | 14.04%     |
| Sunrise Boulevard & Antelope Road     | NNE  | 0.41          | 1,705    | 0.1          | 13.97%     |
| Antelope Road & Old Auburn Road       | ENE  | 0.43          | 1,705    | 0.1          | 13.87%     |
| Old Auburn Road & Sunrise Boulevard   | NNE  | 0.15          | 1,500    | 0.1          | 13.53%     |
| Sungarden Drive & Sunrise Boulevard   | SSE  | 0.44          | 1,500    | 0.1          | 12.20%     |
| Fair Oaks Boulevard & Oak Avenue      | ESE  | 0.59          | 1,529    | 0.1          | 11.81%     |
| Fair Oaks Boulevard & Old Auburn Road | ENE  | 0.70          | 1,000    | 0.1          | 7.44%      |
| Mariposa Avenue & Antelope Road       | WNW  | 0.50          | 802      | 0.1          | 6.38%      |

| Study-area zone                   | Dir. | Distance (mi) | Mass (M) | Gravity term | Trip share |
|-----------------------------------|------|---------------|----------|--------------|------------|
| Old Auburn Road & Mariposa Avenue | WSW  | 0.46          | 480      | 0.0          | 3.88%      |
| Sunrise Boulevard & Locher Way    | SSE  | 0.66          | 380      | 0.0          | 2.87%      |

10 study-area zone(s). Screening-grade distribution; not a calibrated regional model.

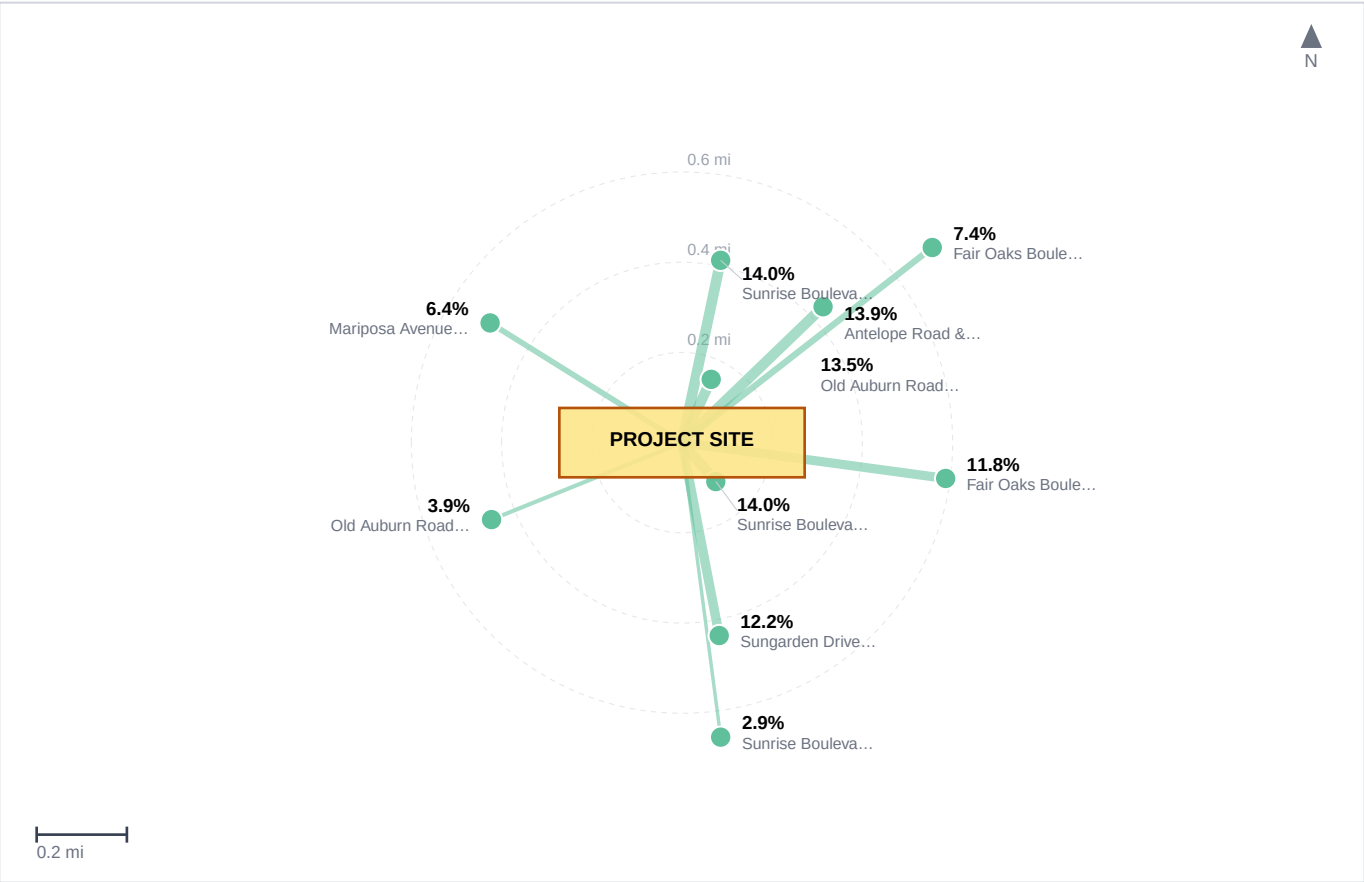
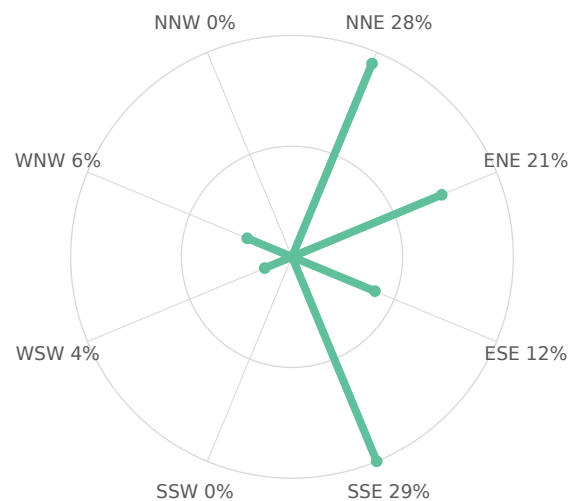


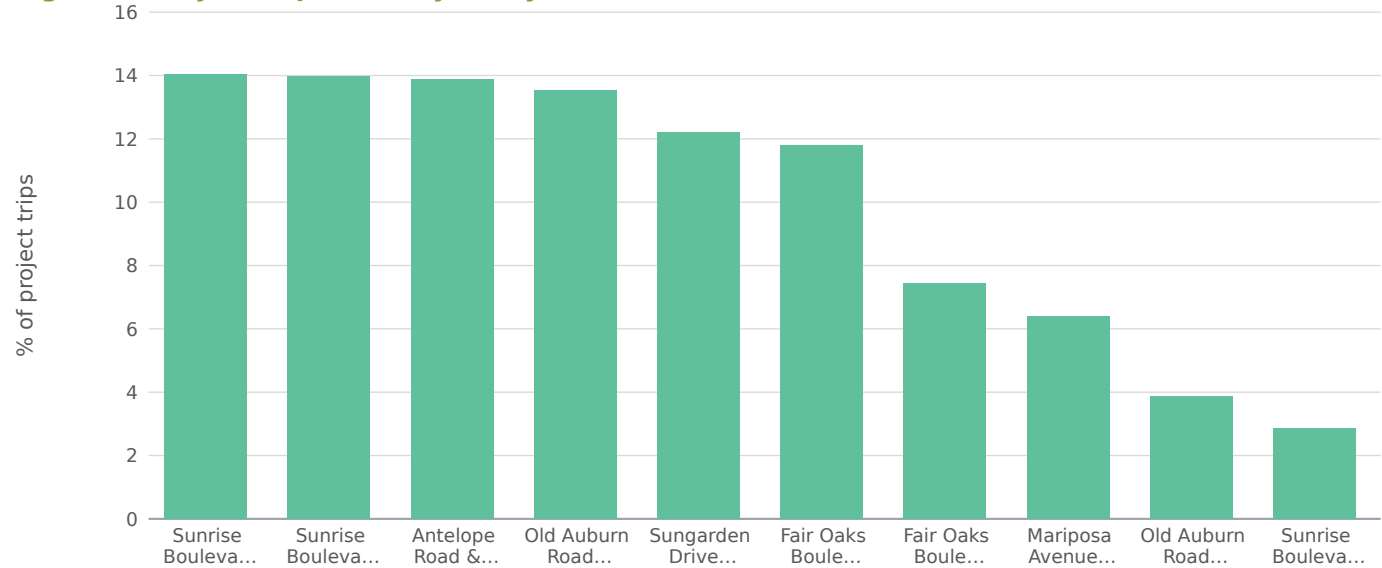
Figure — Project Trip Distribution. Study-area zones plotted to scale at their true bearing and distance from the site; leg weight is proportional to each zone's share of project trips, and the label gives that share. Derived from the Gravity Model distribution — the same shares tabulated above. Screening-grade: zone positions are the analysis locations, not a surveyed base map.

Figure — Directional Distribution of Project Trips



Screening-grade directional distribution of net new project trips by compass octant (spoke length  $\propto$  percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone



**Figure — Trip Share vs. Distance from Site (Gravity Decay)**



#### 4.4 Project Trip Assignment

| Study intersection                    | Dir. | Distance (mi) | AM trips | PM trips | Share of project |
|---------------------------------------|------|---------------|----------|----------|------------------|
| Sunrise Boulevard & Oak Avenue        | SSE  | 0.11          | 52       | 115      | 11.3%            |
| Old Auburn Road & Sunrise Boulevard   | NNE  | 0.15          | 84       | 213      | 20.8%            |
| Sunrise Boulevard & Antelope Road     | NNE  | 0.41          | 30       | 75       | 7.3%             |
| Antelope Road & Old Auburn Road       | ENE  | 0.43          | 48       | 91       | 8.9%             |
| Sungarden Drive & Sunrise Boulevard   | SSE  | 0.44          | 36       | 61       | 6.0%             |
| Old Auburn Road & Mariposa Avenue     | WSW  | 0.46          | 97       | 148      | 14.5%            |
| Mariposa Avenue & Antelope Road       | WNW  | 0.50          | 76       | 113      | 11.1%            |
| Fair Oaks Boulevard & Oak Avenue      | ESE  | 0.59          | 44       | 95       | 9.3%             |
| Sunrise Boulevard & Locher Way        | SSE  | 0.66          | 8        | 20       | 2.0%             |
| Fair Oaks Boulevard & Old Auburn Road | ENE  | 0.70          | 48       | 91       | 8.9%             |

Project trips assigned to each study intersection from the distribution shares above: a four-step gravity model sets the share reaching each intersection from its attraction mass and its distance from the site. Directional load multipliers are a Florida (Caltran) refinement and are not applied in this region — each intersection's loading is taken directly from the gravity weights above.

#### 4.5 Affected Intersections — Existing / Opening / 20-Year Design

Four scenarios are evaluated at each affected intersection per Caltrans / local-jurisdiction operational-analysis convention (LOS-only context, not CEQA-VMT): (1) Existing — current-year volumes, no growth; (2) Opening-Year No-Build (2027) — existing grown at 1.5%/yr; (3) Opening-Year Build — No-Build plus project external trips; (4) 20-Year Design Year (2047) No-Build and Build — opening volumes compounded another 20 years, project trips at full build-out unchanged.

| Intersection                          | Existing | Opening NB | Opening Bld | Design NB | Design Bld | Δ delay (s) |
|---------------------------------------|----------|------------|-------------|-----------|------------|-------------|
| Sunrise Boulevard & Oak Avenue        | C        | C          | ▲ D         | F         | F          | 7.5         |
| Old Auburn Road & Sunrise Boulevard   | C        | C          | ▲ D         | F         | F          | 15.4        |
| Sunrise Boulevard & Antelope Road     | D        | D          | ▲ E         | F         | F          | 9.4         |
| Antelope Road & Old Auburn Road       | D        | D          | ▲ E         | F         | F          | 11.6        |
| Sungarden Drive & Sunrise Boulevard   | C        | C          | ▲ D         | F         | F          | 3.0         |
| Old Auburn Road & Mariposa Avenue     | B        | B          | B           | B         | B          | 1.1         |
| Mariposa Avenue & Antelope Road       | B        | B          | ▲ C         | C         | C          | 1.2         |
| Fair Oaks Boulevard & Oak Avenue      | C        | C          | ▲ D         | F         | F          | 5.9         |
| Sunrise Boulevard & Locher Way        | B        | B          | B           | B         | B          | 0.1         |
| Fair Oaks Boulevard & Old Auburn Road | C        | C          | C           | C         | C          | 1.2         |

#### 4.6 Recommended Operational Improvements (Non-CEQA)

Improvements below address non-CEQA operational impact only. Per § 15064.3, intersection geometry improvements do NOT reduce VMT and are NOT creditable as CEQA mitigation; CEQA mitigation must come from CAPCOA 2024 (see §3.5).

##### **Sunrise Boulevard & Oak Avenue [MAJOR]**

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +7.4s, design year +26.1s).

##### **Old Auburn Road & Sunrise Boulevard [MAJOR]**

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +15.5s, design year +48.5s).

##### **Sunrise Boulevard & Antelope Road [MAJOR]**

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +9.4s, design year +18.0s).

##### **Antelope Road & Old Auburn Road [MAJOR]**

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +11.6s, design year +21.8s).

##### **Sungarden Drive & Sunrise Boulevard [MAJOR]**

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +3.1s, design year +13.4s).

##### **Fair Oaks Boulevard & Oak Avenue [MAJOR]**

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +5.9s, design year +21.5s).

## 5.0 CALTRANS COORDINATION

If the project fronts on a Caltrans State Highway System (SHS) facility or proposes new access to an SHS route, the following Caltrans coordination items apply:

CEQA significance on SHS

Caltrans Transportation Analysis under CEQA (TAC, 2nd Ed., Sept 2025)

Induced-travel analysis

Caltrans Transportation Analysis Framework (TAF, 2nd Ed., Sept 2025) — capacity-increasing SHS projects only

Design references

Caltrans Highway Design Manual (HDM, 7th Edition), Chapters 100 + 400

Encroachment permitting

Caltrans Encroachment Permits Manual (EPM) — non-CEQA, LOS-based operational analysis required

Signal warrants

California MUTCD 2026 (effective 2026-01-18), Part 4C

Signal timing operations

Caltrans Traffic Signal Operations Manual (Jan 2020)

## 6.0 FINDINGS

- Project will generate 6,800 new daily vehicle trips, with 680 during the PM peak hour (326 inbound / 354 outbound).
- Pass-by credit 30% and internal-capture credit 0% applied at the PM peak (25% of that credit at the AM and Saturday-midday periods, per the industry rule of thumb that off-peak shopping diverts less) before off-site assignment (standard pass-by methodology; ULI Internal Capture).
- Auto-mode share of 89% applied to external trips for Sacramento-Roseville-Folsom MSA; 11% of generated trips are assumed to arrive by transit, walking, or cycling and do not load the off-site roadway network. Source: US suburban default (ACS B08301 metropolitan median  $\approx$  90%).
- Existing volumes were grown by 1.50%/yr over 1 year ( $\times 1.01$ ) to the opening-year horizon.
- 10 signalized intersections fall within the study area; 7 are projected to drop at least one LOS grade in the opening year after build-out.
- 2 intersections are projected to operate at LOS E or F under the opening-year build condition and require formal mitigation per Sacramento Department of Public Works TIS guidance.
- Worst opening-year impact: Old Auburn Road & Sunrise Boulevard — projected delay rises 15.4s (LOS C  $\rightarrow$  D); 95th-pct queue 488 ft on the critical approach.
- Largest projected delay change: 15.4s in the opening year (Old Auburn Road & Sunrise Boulevard) and 48.5s in the design year (Old Auburn Road & Sunrise Boulevard). The design-year figure is the larger; that difference is driven by background traffic growth over the design horizon, not by the development.
- Conclusions are reported at the applied screening assumptions. Discrete-scenario sensitivity at the formal TIA scoping meeting should test: trip-generation method (rate vs. fitted-curve equation); internal capture and pass-by credit variants; and a  $\pm 0.5\%$ /yr background growth band around the applied value.

## 7.0 METHODOLOGY NOTES (NON-CEQA OPERATIONAL)

- Trip generation uses public-data average rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) for the selected land-use code, computed for AM peak, PM peak, Saturday midday, and daily totals. Saturday-midday rates are estimated as a published industry multiple of the PM peak rate by land-use category.
- Pass-by and internal-capture credits are applied in full at the PM peak (and at 25% of that credit fraction for the AM and Saturday-midday periods) per standard pass-by screening methodology and ULI Mixed-Use Internal Capture defaults; only the residual external vehicle trips are assigned to off-site intersections.
- Existing intersection volumes are grown to the opening-year horizon at the user-supplied annual growth rate (default 1.5%/yr) before the capacity analysis.
- Weather adjustment applies published rain/snow capacity-reduction factors as adopted in US traffic-engineering practice: clear 1.00, light rain 0.95, heavy rain 0.86, light snow 0.86, heavy snow 0.70. The factor multiplies the saturation flow at every intersection.
- Off-site impact is screened for all signalized intersections within the study radius (default 0.5 mi) using the four-step travel demand model (FHWA; NCHRP Report 716). Step 1 Trip Generation: public-data average rates (SANDAG 2002 / NHTS 2017 / NCHRP 716) give the site's external (post pass-by / internal-capture) productions. Step 2 Trip Distribution: a production-constrained gravity model  $T_j = P \cdot (A_j \cdot F_j) / \sum (A_x \cdot F_x)$  allocates trips to surrounding signals, where attractiveness  $A_j$  is the signal's through-volume and the friction factor  $F_j$  is the NCHRP-716 gamma function  $F = a \cdot t^b \cdot e^{-(c \cdot t)}$  (home-based-work coefficients  $a=28507$ ,  $b=-0.02$ ,  $c=-0.123$ ) on the travel time  $t$  to each signal. Step 3 Mode Choice: a binary logit  $P(\text{auto}) = 1 / (1 + e^{-(\text{ASC} - \lambda \cdot \Delta \text{GC})})$  calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (a density proxy from surrounding through-volumes) so denser, more transit-served sites split further from auto; only the resulting vehicle trips load the network. Step 4 Route Assignment: a capacity-constrained assignment using the BPR

volume-delay function  $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$  iteratively shifts trips away from over-capacity signals toward less-congested alternatives. Signals lacking AADT data fall back to a constant 5,000 vpd attraction.

- After assignment, all signalized intersections within the study radius are reported in the affected-intersections table. Project-added trips, v/c ratio change, control delay change, LOS change, and 95th-percentile queue are reported for each intersection so the reviewer can assess relative impact. Intersections beyond the study radius are excluded from analysis.

- Auto-mode share is applied per metro before assignment. Suburban-US metros default to 90% auto (ACS 5-Year B08301 median); transit-heavy metros use measured auto-mode share (e.g., NYC 32%, Tokyo 30%, London 38%, San Francisco 47%). Non-auto trips (transit, walking, cycling) do not load the off-site roadway. This is a screening-level adjustment; a real TIS submittal in a transit-heavy market should refine with project-specific TAZ data.

- Candidate signals are de-duplicated within a 45m clustering threshold to prevent OSM divided-arterial splits and way-record artifacts from double-counting a single physical intersection.

- Intersection-level control delay uses the standard signalized-intersection model  $d = d_1 + d_2$  (Webster uniform delay + Akçelik time-dependent incremental-delay term, both openly published) with a 90s cycle,  $g/C = 0.45$ , 1,800 vphpl saturation flow ( $\times$  weather factor), 15-minute peak analysis period ( $T = 0.25$  hr) and pretimed-signal incremental-delay factor  $k = 0.5$ . Reported control delay is capped at 300 s (LOS F) for screening reliability — the incremental-delay term is not calibrated far above capacity, so oversaturated approaches are reported as LOS F rather than an implausible delay figure; a calibrated design-level analysis (HCS / Synchro) supersedes this screen.

- Approach-level analysis splits each signal's inflow across NB/SB/EB/WB approaches (deterministic per-signal allocation perturbed  $\pm 15\%$  from a 30/25/25/20 base) and assigns added trips to each approach by cosine-similarity to the bearing of the project relative to the signal. Per-approach v/c, control delay, LOS, and 95th-percentile back-of-queue length (standard cyclic-queue relation with Poisson percentile factor:  $Q_{95} \approx Q_1 \times 1.65 \times 25 \text{ ft/veh}$ ) are reported.

- Level of Service is assigned from the conventional US signalized control-delay bands republished in state DOT traffic-engineering manuals: A  $\leq 10s$ , B  $\leq 20s$ , C  $\leq 35s$ , D  $\leq 55s$ , E  $\leq 80s$ , F  $> 80s$ .

- Sensitivity is reported in narrative form per standard TIA practice: trip-generation method (rate vs. fitted-curve equation), discrete internal capture and pass-by credit variants, and a  $\pm 0.5\%/yr$  growth-rate band around the applied value. The engine retains an internal stochastic-sensitivity routine (Box-Muller-perturbed trip rate and existing volume) for demo-mode diagnostics; it is not surfaced in the deliverable because TIA sensitivity is conducted through discrete scoping-agreed scenario variants, not statistical perturbation of unmeasured distributions.

- Mitigations are screening-level recommendations sized to the projected delay change, not full Synchro/SimTraffic optimization runs. A formal TIS submittal should validate these recommendations with detailed traffic counts and signal-timing analysis.

- Turbo-lane (continuous-green-T) screening flags signalized 3-leg T-intersections where one or more main-street through lanes could flow continuously while the minor-street left turn merges in the median — recovering the green time the through movement would otherwise lose to the signal. Candidacy requires measured 3-leg geometry, an arterial-class main street, and a detected median (divided carriageway), all derived from the OpenStreetMap road network. Approach-capacity gain is computed as  $(\text{turbo lanes} / \text{approach lanes}) \times (1 - g/C) / (g/C)$ , with the main-street through  $g/C$  derived from the modeled main-vs-minor critical-flow split; the result is reported within the  $+7\% \dots +173\%$  envelope documented in the continuous-green-T design literature — the standard national references for this geometry (David Plummer & Associates, Adding Turbo Lanes to T-Intersections, 2010; and the Design Guidelines for the Development of Continuous Green Intersections, 1997). Lane counts and signal timing must be field-verified before design.



## FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

### Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

|                           |                                   |
|---------------------------|-----------------------------------|
| Land use                  | LU 820 — Shopping Center / Retail |
| Size                      | 85 1,000 sqft GLA                 |
| Daily trips (productions) | 6,800                             |
| PM peak-hour trips        | 680 (326 in / 354 out)            |

### Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals:  $T_j = P \cdot (A_j \cdot F_j) / \sum (A_x \cdot F_x)$ . Attractiveness  $A_j$  is each signal's through-volume; the friction factor  $F_j$  is a gamma function  $F = a \cdot t^b \cdot e^{-(c \cdot t)}$  — the form recommended by NCHRP Report 716 — on the travel time  $t$  to the signal, with home-based-work shape parameters  $b=-0.02$ ,  $c=-0.123$  (consistent with NCHRP 716's sampled MPO gravity parameters; the scale constant  $a$  cancels in the share normalization).

| Signal (attraction zone)              | Dist mi | Time min | PM trips | Share |
|---------------------------------------|---------|----------|----------|-------|
| Old Auburn Road & Sunrise Boulevard   | 0.15    | 0.4      | 213      | 20.8% |
| Old Auburn Road & Mariposa Avenue     | 0.46    | 1.1      | 148      | 14.5% |
| Sunrise Boulevard & Oak Avenue        | 0.11    | 0.3      | 115      | 11.3% |
| Mariposa Avenue & Antelope Road       | 0.50    | 1.2      | 113      | 11.1% |
| Fair Oaks Boulevard & Oak Avenue      | 0.59    | 1.4      | 95       | 9.3%  |
| Antelope Road & Old Auburn Road       | 0.43    | 1.0      | 91       | 8.9%  |
| Fair Oaks Boulevard & Old Auburn Road | 0.70    | 1.7      | 91       | 8.9%  |
| Sunrise Boulevard & Antelope Road     | 0.41    | 1.0      | 75       | 7.3%  |
| Sungarden Drive & Sunrise Boulevard   | 0.44    | 1.1      | 61       | 6.0%  |
| Sunrise Boulevard & Locher Way        | 0.66    | 1.6      | 20       | 2.0%  |

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

### Step 3 — Mode Choice

Auto-mode share 89% applied via a binary logit  $P(\text{auto}) = 1 / (1 + e^{-(\text{ASC} - \lambda \cdot \Delta \text{GC})})$  calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 11% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

### Step 4 — Route Assignment

Project trips are routed over the local road network by shortest path and loaded with a capacity-constrained BPR equilibrium (volume-delay  $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$ , Method of Successive Averages), so congested links shift trips toward alternative routes. 100% of study signals lie on a modeled route; highest loaded-link  $v/c$ : 0.37. Corridors carrying the project trips:

| Corridor (road class) | Loaded length mi | PM project vph | v/c  |
|-----------------------|------------------|----------------|------|
| Collector             | 0.10             | 426            | 0.37 |
| Minor arterial        | 1.66             | 409            | 0.36 |
| Principal arterial    | 1.63             | 174            | 0.24 |

Network shortest-path assignment over the OSM road graph within the study area; per-signal v/c, delay, LOS, and queue are in the capacity appendix.

Conserved assignment: project trips routed to 18 cordon gateways on the study boundary (weighted by the directional distribution above). Flow conservation verified at render input:  $\Sigma$  entering =  $\Sigma$  leaving at 366 network junctions, max imbalance 0.00 veh. 10 study intersection(s) carry path-derived movements; 0 retain the geometric octant model (labeled per intersection).

## APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

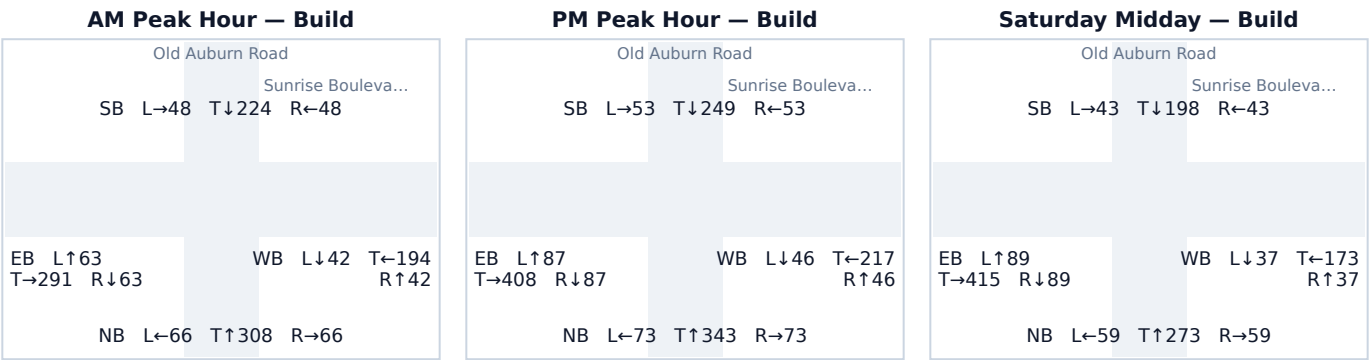
One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach capacity table. Analysis uses the openly-published Webster/Akcelik signalized control-delay method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 Old Auburn Road & Sunrise Boulevard

|                          |                                                                                                                                                                                                                                                |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signal ID                | sacramento-3664                                                                                                                                                                                                                                |
| Location                 | NE Sacramento-Roseville-Folsom · 0.15 mi from site                                                                                                                                                                                             |
| Added PM peak trips      | 213                                                                                                                                                                                                                                            |
| Existing / No-Build (PM) | LOS C · 32.6 s/veh · v/c 0.85                                                                                                                                                                                                                  |
| Build (PM)               | LOS D · 48.0 s/veh · v/c 0.96                                                                                                                                                                                                                  |
| Δ control delay          | +15.4 s/veh (LOS change)                                                                                                                                                                                                                       |
| Mitigation               | Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +15.5s, design year +48.5s). |

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

| Appr. | NB vph | +Trips | Build vph | v/c NB→Bld  | Delay NB→Bld | LOS NB→Bld | Q95 ft |
|-------|--------|--------|-----------|-------------|--------------|------------|--------|
| NB    | 489    | 0      | 489       | 0.60 → 0.60 | 22.0 → 22.0  | C → C      | 381    |
| SB    | 355    | 0      | 355       | 0.44 → 0.44 | 18.7 → 18.7  | B → B      | 251    |
| EB    | 369    | 213    | 582       | 0.46 → 0.72 | 19.0 → 25.6  | B → C      | 488    |
| WB    | 309    | 0      | 309       | 0.38 → 0.38 | 17.8 → 17.8  | B → B      | 212    |

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

| Movement   | Project trips | % of project trips |
|------------|---------------|--------------------|
| EB Through | 122           | 57%                |
| EB Right   | 91            | 43%                |

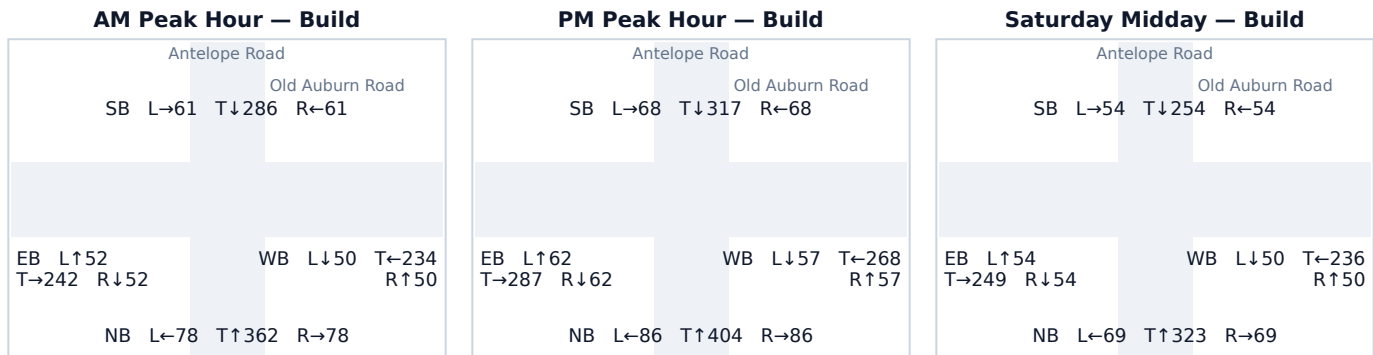
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

## A.2 Antelope Road & Old Auburn Road

|                          |                                                                                                                                                                                                                                                |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signal ID                | sacramento-544                                                                                                                                                                                                                                 |
| Location                 | NE Sacramento-Roseville-Folsom · 0.43 mi from site 91                                                                                                                                                                                          |
| Added PM peak trips      | LOS D · 47.5 s/veh · v/c 0.96                                                                                                                                                                                                                  |
| Existing / No-Build (PM) | LOS E · 59.1 s/veh · v/c 1.01                                                                                                                                                                                                                  |
| Build (PM)               | +11.6 s/veh (LOS change)                                                                                                                                                                                                                       |
| Δ control delay          | Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +11.6s, design year +21.8s). |
| Mitigation               |                                                                                                                                                                                                                                                |

### Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

### Per-Approach Capacity (PM Peak)

| Appr. | NB vph | +Trips | Build vph | v/c NB→Bld  | Delay NB→Bld | LOS NB→Bld | Q95 ft |
|-------|--------|--------|-----------|-------------|--------------|------------|--------|
| NB    | 576    | 0      | 576       | 0.71 → 0.71 | 25.3 → 25.3  | C → C      | 480    |
| SB    | 453    | 0      | 453       | 0.56 → 0.56 | 21.0 → 21.0  | C → C      | 343    |
| EB    | 364    | 47     | 411       | 0.45 → 0.51 | 18.9 → 19.9  | B → B      | 302    |
| WB    | 338    | 44     | 382       | 0.42 → 0.47 | 18.3 → 19.2  | B → B      | 275    |

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

### Affected movements (PM peak project trips)

| Movement   | Project trips | % of project trips |
|------------|---------------|--------------------|
| EB Through | 47            | 52%                |
| WB Right   | 44            | 48%                |

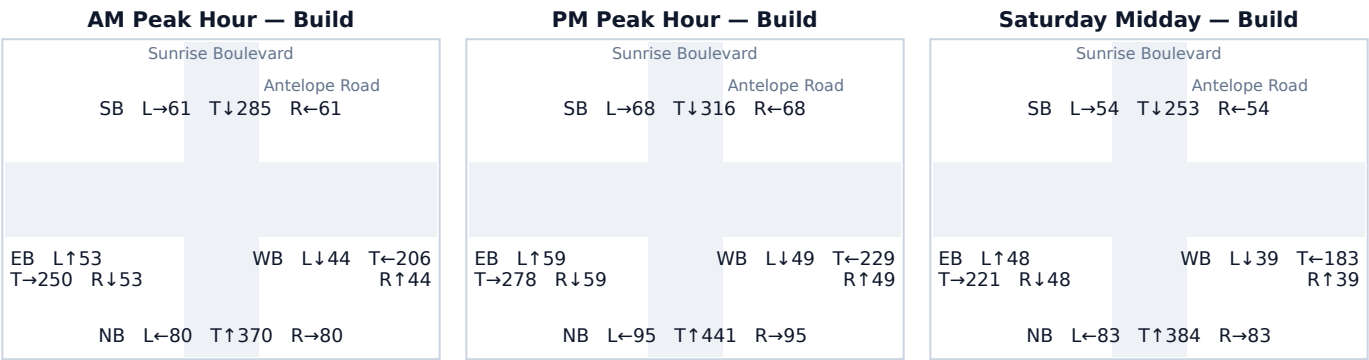
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.3 Sunrise Boulevard & Antelope Road

|                          |                                                                                                                                                                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signal ID                | sacramento-3661                                                                                                                                                                                                                               |
| Location                 | NE Sacramento-Roseville-Folsom · 0.41 mi from site                                                                                                                                                                                            |
| Added PM peak trips      | 75                                                                                                                                                                                                                                            |
| Existing / No-Build (PM) | LOS D · 47.5 s/veh · v/c 0.96                                                                                                                                                                                                                 |
| Build (PM)               | LOS E · 56.9 s/veh · v/c 1.00                                                                                                                                                                                                                 |
| Δ control delay          | +9.4 s/veh (LOS change)                                                                                                                                                                                                                       |
| Mitigation               | Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +9.4s, design year +18.0s). |

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

| Appr. | NB vph | +Trips | Build vph | v/c NB→Bld  | Delay NB→Bld | LOS NB→Bld | Q95 ft |
|-------|--------|--------|-----------|-------------|--------------|------------|--------|
| NB    | 556    | 75     | 631       | 0.69 → 0.78 | 24.4 → 28.3  | C → C      | 552    |
| SB    | 452    | 0      | 452       | 0.56 → 0.56 | 20.9 → 20.9  | C → C      | 342    |
| EB    | 396    | 0      | 396       | 0.49 → 0.49 | 19.6 → 19.6  | B → B      | 288    |
| WB    | 327    | 0      | 327       | 0.40 → 0.40 | 18.1 → 18.1  | B → B      | 226    |

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

| Movement   | Project trips | % of project trips |
|------------|---------------|--------------------|
| NB Through | 75            | 100%               |

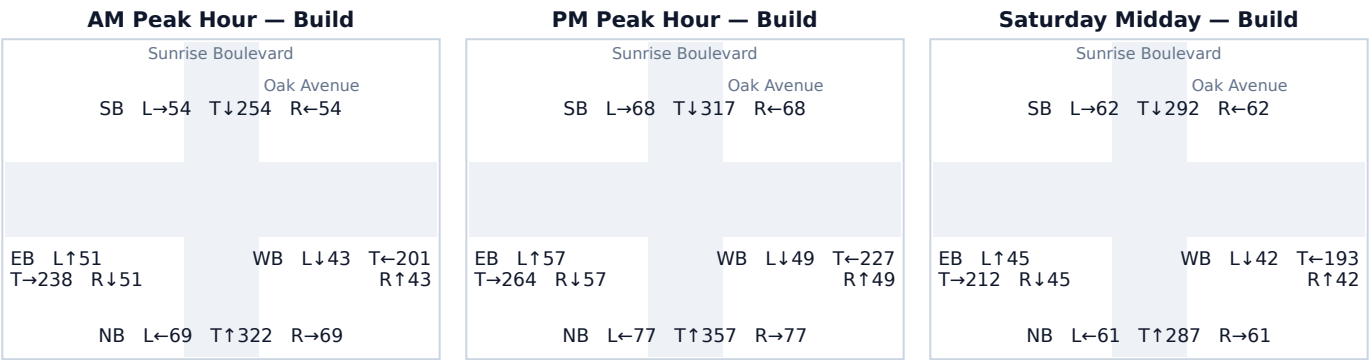
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.4 Sunrise Boulevard & Oak Avenue

|                          |                                                                                                                                                                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signal ID                | sacramento-2809                                                                                                                                                                                                                               |
| Location                 | NE Sacramento-Roseville-Folsom · 0.11 mi from site 115                                                                                                                                                                                        |
| Added PM peak trips      | LOS C · 33.9 s/veh · v/c 0.86                                                                                                                                                                                                                 |
| Existing / No-Build (PM) | LOS D · 41.4 s/veh · v/c 0.93                                                                                                                                                                                                                 |
| Build (PM)               | +7.5 s/veh (LOS change)                                                                                                                                                                                                                       |
| Δ control delay          | Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +7.4s, design year +26.1s). |
| Mitigation               |                                                                                                                                                                                                                                               |

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

| Appr. | NB vph | +Trips | Build vph | v/c NB→Bld  | Delay NB→Bld | LOS NB→Bld | Q95 ft |
|-------|--------|--------|-----------|-------------|--------------|------------|--------|
| NB    | 511    | 0      | 511       | 0.63 → 0.63 | 22.7 → 22.7  | C → C      | 405    |
| SB    | 362    | 91     | 453       | 0.45 → 0.56 | 18.8 → 21.0  | B → C      | 343    |
| EB    | 378    | 0      | 378       | 0.47 → 0.47 | 19.2 → 19.2  | B → B      | 271    |
| WB    | 300    | 24     | 325       | 0.37 → 0.40 | 17.6 → 18.1  | B → B      | 225    |

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

| Movement   | Project trips | % of project trips |
|------------|---------------|--------------------|
| SB Left    | 71            | 62%                |
| WB Left    | 24            | 21%                |
| SB Through | 20            | 17%                |

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split,

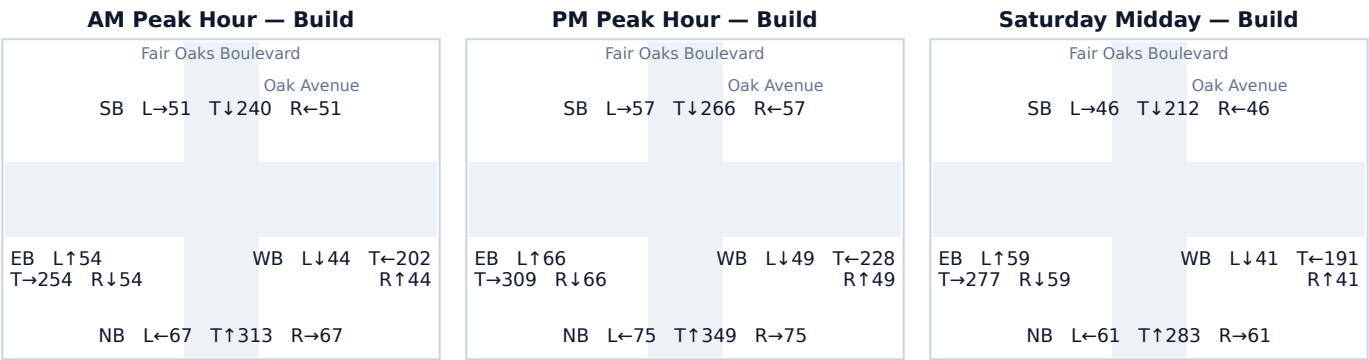


not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.5 Fair Oaks Boulevard & Oak Avenue

|                          |                                                                                                                                                                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signal ID                | sacramento-367                                                                                                                                                                                                                                |
| Location                 | NE Sacramento-Roseville-Folsom · 0.59 mi from site                                                                                                                                                                                            |
| Added PM peak trips      | 95                                                                                                                                                                                                                                            |
| Existing / No-Build (PM) | LOS C · 33.9 s/veh · v/c 0.86                                                                                                                                                                                                                 |
| Build (PM)               | LOS D · 39.8 s/veh · v/c 0.91                                                                                                                                                                                                                 |
| Δ control delay          | +5.9 s/veh (LOS change)                                                                                                                                                                                                                       |
| Mitigation               | Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +5.9s, design year +21.5s). |

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

| Appr. | NB vph | +Trips | Build vph | v/c NB→Bld  | Delay NB→Bld | LOS NB→Bld | Q95 ft |
|-------|--------|--------|-----------|-------------|--------------|------------|--------|
| NB    | 491    | 8      | 499       | 0.61 → 0.62 | 22.1 → 22.3  | C → C      | 391    |
| SB    | 380    | 0      | 380       | 0.47 → 0.47 | 19.2 → 19.2  | B → B      | 274    |
| EB    | 371    | 71     | 441       | 0.46 → 0.54 | 19.0 → 20.7  | B → C      | 332    |
| WB    | 310    | 16     | 326       | 0.38 → 0.40 | 17.8 → 18.1  | B → B      | 226    |

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

| Movement   | Project trips | % of project trips |
|------------|---------------|--------------------|
| EB Right   | 53            | 56%                |
| EB Through | 18            | 19%                |
| WB Through | 16            | 17%                |
| NB Left    | 8             | 8%                 |

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

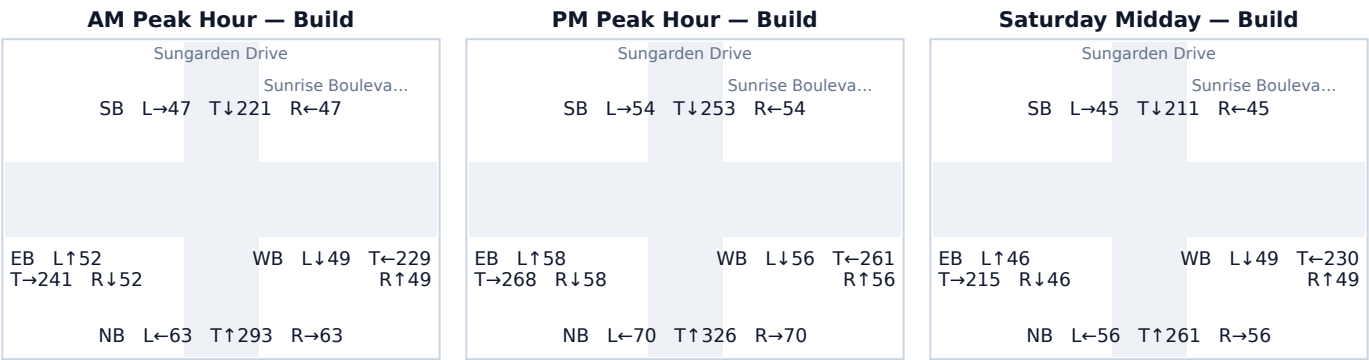
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-

foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.6 Sungarden Drive & Sunrise Boulevard

|                          |                                                                                                                                                                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signal ID                | sacramento-2806                                                                                                                                                                                                                               |
| Location                 | NE Sacramento-Roseville-Folsom · 0.44 mi from site                                                                                                                                                                                            |
| Added PM peak trips      | 61                                                                                                                                                                                                                                            |
| Existing / No-Build (PM) | LOS C · 32.6 s/veh · v/c 0.85                                                                                                                                                                                                                 |
| Build (PM)               | LOS D · 35.6 s/veh · v/c 0.88                                                                                                                                                                                                                 |
| Δ control delay          | +3.0 s/veh (LOS change)                                                                                                                                                                                                                       |
| Mitigation               | Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +3.1s, design year +13.4s). |

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

| Appr. | NB vph | +Trips | Build vph | v/c NB→Bld  | Delay NB→Bld | LOS NB→Bld | Q95 ft |
|-------|--------|--------|-----------|-------------|--------------|------------|--------|
| NB    | 466    | 0      | 466       | 0.58 → 0.58 | 21.3 → 21.3  | C → C      | 356    |
| SB    | 341    | 20     | 361       | 0.42 → 0.45 | 18.4 → 18.8  | B → B      | 256    |
| EB    | 384    | 0      | 384       | 0.47 → 0.47 | 19.3 → 19.3  | B → B      | 277    |
| WB    | 332    | 41     | 373       | 0.41 → 0.46 | 18.2 → 19.1  | B → B      | 267    |

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

| Movement   | Project trips | % of project trips |
|------------|---------------|--------------------|
| WB Through | 41            | 67%                |
| SB Through | 20            | 33%                |

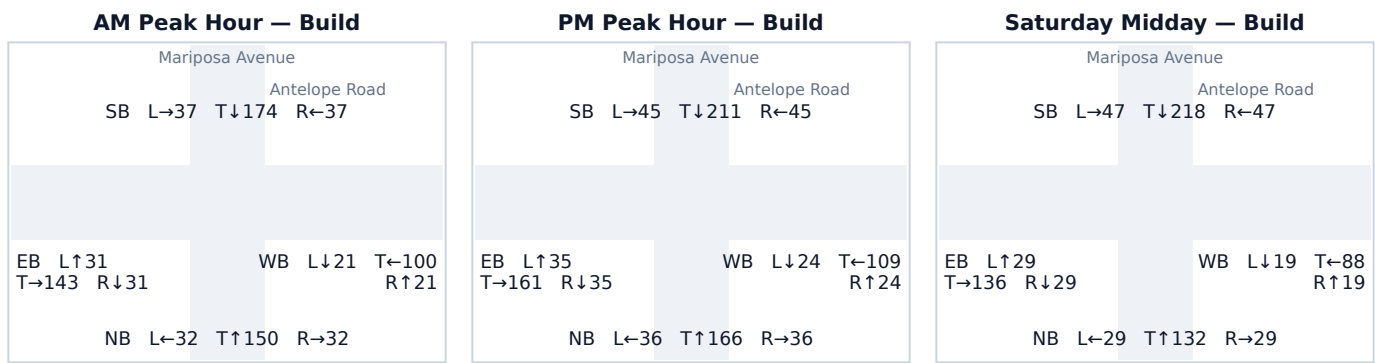
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

## A.7 Mariposa Avenue & Antelope Road

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signal ID                | sacramento-3658                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Location                 | NE Sacramento-Roseville-Folsom · 0.50 mi from site                                                                                                                                                                                                                                                                                                                                                                                            |
| Added PM peak trips      | 113                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Existing / No-Build (PM) | LOS B · 18.9 s/veh · v/c 0.45                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Build (PM)               | LOS C · 20.1 s/veh · v/c 0.51                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Δ control delay          | +1.2 s/veh (LOS change)                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Mitigation               | No mitigation indicated at screening level (opening year +1.1s, design year +1.8s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here. |

### Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

### Per-Approach Capacity (PM Peak)

| Appr. | NB vph | +Trips | Build vph | v/c NB→Bld  | Delay NB→Bld | LOS NB→Bld | Q95 ft |
|-------|--------|--------|-----------|-------------|--------------|------------|--------|
| NB    | 238    | 0      | 238       | 0.29 → 0.29 | 16.6 → 16.6  | B → B      | 155    |
| SB    | 201    | 100    | 301       | 0.25 → 0.37 | 16.1 → 17.7  | B → B      | 205    |
| EB    | 218    | 13     | 231       | 0.27 → 0.29 | 16.3 → 16.5  | B → B      | 150    |
| WB    | 157    | 0      | 157       | 0.19 → 0.19 | 15.5 → 15.5  | B → B      | 98     |

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

### Affected movements (PM peak project trips)

| Movement   | Project trips | % of project trips |
|------------|---------------|--------------------|
| SB Through | 100           | 88%                |
| EB Right   | 13            | 12%                |

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

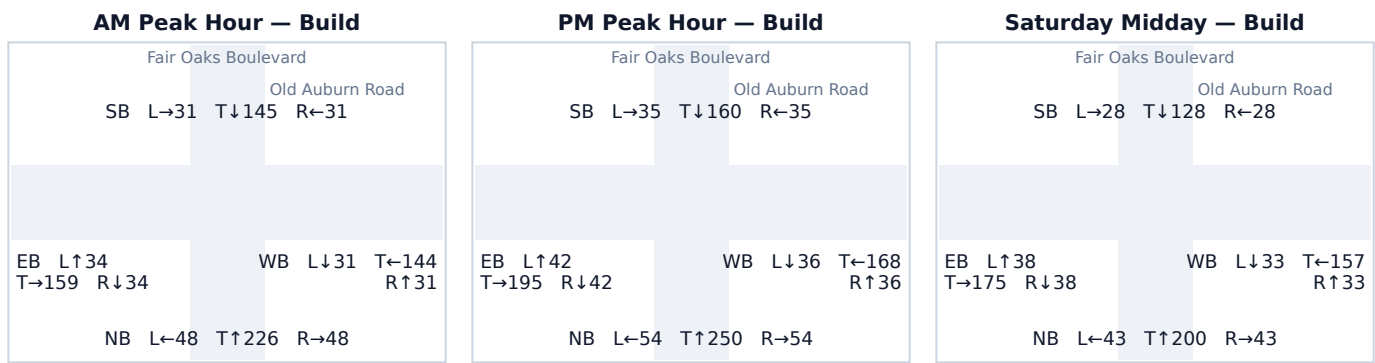
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-

foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

## A.8 Fair Oaks Boulevard & Old Auburn Road

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signal ID                | sacramento-3229                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Location                 | NE Sacramento-Roseville-Folsom · 0.70 mi from site 91                                                                                                                                                                                                                                                                                                                                                                                         |
| Added PM peak trips      | LOS C · 21.1 s/veh · v/c 0.56                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Existing / No-Build (PM) | LOS C · 22.3 s/veh · v/c 0.61                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Build (PM)               | +1.2 s/veh                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Δ control delay          | No mitigation indicated at screening level (opening year +1.2s, design year +2.7s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here. |
| Mitigation               |                                                                                                                                                                                                                                                                                                                                                                                                                                               |

### Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

### Per-Approach Capacity (PM Peak)

| Appr. | NB vph | +Trips | Build vph | v/c NB→Bld  | Delay NB→Bld | LOS NB→Bld | Q95 ft |
|-------|--------|--------|-----------|-------------|--------------|------------|--------|
| NB    | 358    | 0      | 358       | 0.44 → 0.44 | 18.7 → 18.7  | B → B      | 253    |
| SB    | 230    | 0      | 230       | 0.28 → 0.28 | 16.5 → 16.5  | B → B      | 150    |
| EB    | 231    | 47     | 279       | 0.29 → 0.34 | 16.5 → 17.3  | B → B      | 187    |
| WB    | 196    | 44     | 240       | 0.24 → 0.30 | 16.0 → 16.6  | B → B      | 157    |

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

### Affected movements (PM peak project trips)

| Movement   | Project trips | % of project trips |
|------------|---------------|--------------------|
| EB Through | 47            | 52%                |
| WB Through | 44            | 48%                |

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-

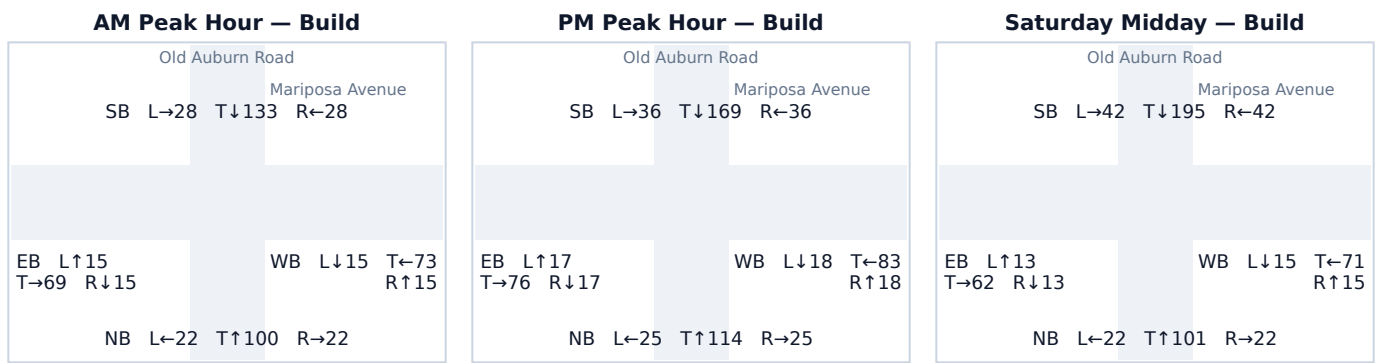
foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.



## A.9 Old Auburn Road & Mariposa Avenue

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signal ID                | sacramento-254                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Location                 | NE Sacramento-Roseville-Folsom · 0.46 mi from site 148                                                                                                                                                                                                                                                                                                                                                                                        |
| Added PM peak trips      | LOS B · 16.3 s/veh · v/c 0.27                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Existing / No-Build (PM) | LOS B · 17.4 s/veh · v/c 0.35                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Build (PM)               | +1.1 s/veh                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Δ control delay          | No mitigation indicated at screening level (opening year +1.1s, design year +1.3s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here. |
| Mitigation               |                                                                                                                                                                                                                                                                                                                                                                                                                                               |

### Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

### Per-Approach Capacity (PM Peak)

| Appr. | NB vph | +Trips | Build vph | v/c NB→Bld  | Delay NB→Bld | LOS NB→Bld | Q95 ft |
|-------|--------|--------|-----------|-------------|--------------|------------|--------|
| NB    | 146    | 18     | 164       | 0.18 → 0.20 | 15.3 → 15.5  | B → B      | 103    |
| SB    | 120    | 121    | 241       | 0.15 → 0.30 | 15.0 → 16.7  | B → B      | 158    |
| EB    | 110    | 0      | 110       | 0.14 → 0.14 | 14.8 → 14.8  | B → B      | 67     |
| WB    | 111    | 9      | 119       | 0.14 → 0.15 | 14.9 → 15.0  | B → B      | 73     |

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

### Affected movements (PM peak project trips)

| Movement | Project trips | % of project trips |
|----------|---------------|--------------------|
| SB Left  | 121           | 82%                |
| NB Right | 18            | 12%                |

| Movement   | Project trips | % of project trips |
|------------|---------------|--------------------|
| WB Through | 9             | 6%                 |

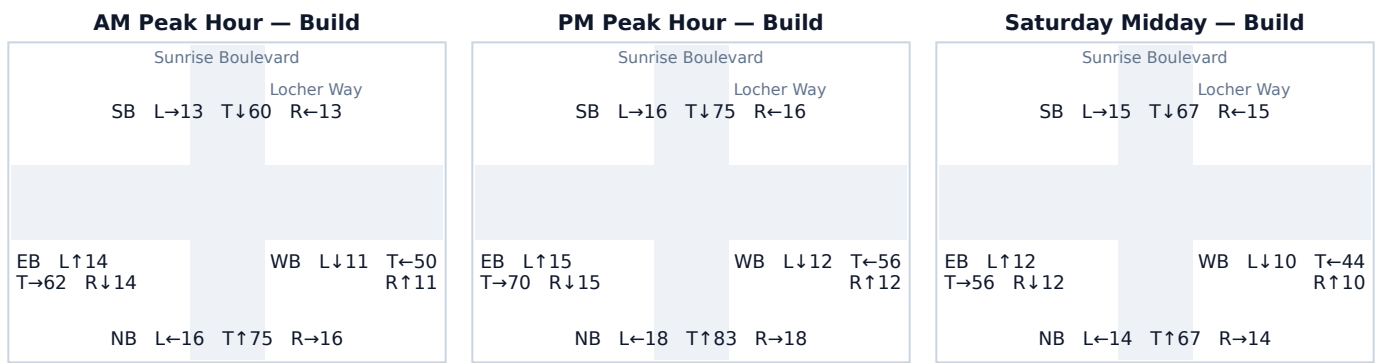
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

## A.10 Sunrise Boulevard & Locher Way

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signal ID                | sacramento-635                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Location                 | NE Sacramento-Roseville-Folsom · 0.66 mi from site                                                                                                                                                                                                                                                                                                                                                                                            |
| Added PM peak trips      | 20                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Existing / No-Build (PM) | LOS B · 15.7 s/veh · v/c 0.21                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Build (PM)               | LOS B · 15.8 s/veh · v/c 0.23                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Δ control delay          | +0.1 s/veh                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Mitigation               | No mitigation indicated at screening level (opening year +0.1s, design year +0.1s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here. |

### Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

### Per-Approach Capacity (PM Peak)

| Appr. | NB vph | +Trips | Build vph | v/c NB→Bld  | Delay NB→Bld | LOS NB→Bld | Q95 ft |
|-------|--------|--------|-----------|-------------|--------------|------------|--------|
| NB    | 119    | 0      | 119       | 0.15 → 0.15 | 15.0 → 15.0  | B → B      | 72     |
| SB    | 87     | 20     | 107       | 0.11 → 0.13 | 14.6 → 14.8  | B → B      | 64     |
| EB    | 100    | 0      | 100       | 0.12 → 0.12 | 14.7 → 14.7  | B → B      | 60     |
| WB    | 80     | 0      | 80        | 0.10 → 0.10 | 14.5 → 14.5  | B → B      | 48     |

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

### Affected movements (PM peak project trips)

| Movement   | Project trips | % of project trips |
|------------|---------------|--------------------|
| SB Through | 20            | 100%               |

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split,

not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.